Five Additional North Indian Ocean Cyclone Dossiers

Structured best-track research companion for the existing Biparjoy (2023) and Amphan (2020) dossier.

Data note: This is a research/documentation layer. For exact timestamp-level position, wind and pressure values, use the official IMD/RSMC best-track reports and NOAA/NCEI IBTrACS. Narrative summaries below intentionally do not fabricate missing rows.

Selected cyclone comparison

Storm	Year	Basin	Category	Peak wind	Track
FANI	2019	Bay of Bengal	Extremely Severe Cyclonic Storm (ESCS)	115 kt (~213 km/h) estimated peak; IMD operational reports comm	3030 km
VAYU	2019	Arabian Sea	Very Severe Cyclonic Storm (VSCS)	80 kt (~148 km/h), gusting to 165 km/h	1862 km
TAUKTAE	2021	Arabian Sea	Extremely Severe Cyclonic Storm (ESCS)	100 kt (~185 km/h), gusting to 210 km/h	1880 km
YAAS	2021	Bay of Bengal	Very Severe Cyclonic Storm (VSCS)	75–85 kt (~140–157 km/h)	About 703 km
MOCHA	2023	Bay of Bengal	Extremely Severe Cyclonic Storm (ESCS)	110 kt (~204 km/h)	Approx. 2000 km class

1. Cyclone FANI (2019)

Period: 26 April – 4 May 2019
Basin: Bay of Bengal
Classification: Extremely Severe Cyclonic Storm (ESCS)

Parameter	Details
Genesis	East Equatorial Indian Ocean and adjoining southeast Bay of Bengal
Landfall / closest approach	Near Puri, Odisha, India; landfall process began around 0800 IST on 3 May
Peak intensity	115 kt (~213 km/h) estimated peak; IMD operational reports commonly expressed the peak range as 200–210 km/
Minimum central pressure	932 hPa
Track length	3030 km
Life period	About 204–219 hours depending on the exact D-to-D convention/source table

Research significance

Long-lived, very long-track Bay of Bengal cyclone with strong intensification before landfall.

Impact summary

Severe wind and heavy rainfall affected Odisha, West Bengal, Bangladesh and parts of northeast India. Odisha experienced major infrastructure, power and housing damage; large-scale evacuation was carried out before landfall.

ML / structured-data relevance

Useful for studying long-track prediction, rapid intensification, landfall error, and post-landfall weakening.

Meteorological timeline

Date / phase	Meteorological development
25 Apr	Low-pressure area formed over the east Equatorial Indian Ocean and adjoining southeast Bay of Bengal.
26 Apr	System concentrated into a Depression.
27 Apr	Deep Depression and then Cyclonic Storm FANI.
29 Apr	Intensified into a Severe Cyclonic Storm.
30 Apr	Reached Very Severe Cyclonic Storm intensity and continued strengthening.
2 May	Reached peak intensity around 115 kt with very low central pressure near 932 hPa.
3 May	Made landfall near Puri, Odisha; strong winds and heavy rain accompanied the landfall.
4 May	Weakened rapidly inland toward Bangladesh/Gangetic West Bengal and lost organized cyclonic intensity.

Structured-data record

Field	Value
storm_name	FANI
season	2019
basin_code	NI
track_start	26 Apr 2019

Field	Value
track_end	4 May 2019
peak_wind_kt	115
min_pressure_hpa	932
track_length_km	3030

Source set

• IMD/RSMC New Delhi — Preliminary Report on ESCS FANI (2019)

• IMD Annual Report 2019

• NOAA/NCEI IBTrACS Version 4.01 documentation

2. Cyclone VAYU (2019)

Period: 10 – 17 June 2019
Basin: Arabian Sea
Classification: Very Severe Cyclonic Storm (VSCS)

Parameter	Details
Genesis	Southeast Arabian Sea and adjoining Lakshadweep/eastcentral Arabian Sea
Landfall / closest approach	No landfall; skirted the south Gujarat coast and later weakened over the sea
Peak intensity	80 kt (~148 km/h), gusting to 165 km/h
Minimum central pressure	970 hPa
Track length	1862 km
Life period	About 129 hours at D-to-D stage convention

Research significance

A recurving Arabian Sea cyclone that demonstrated how forecast tracks can change as steering conditions evolve.

Impact summary

Strong winds, rainfall and rough seas affected Lakshadweep and the Gujarat coastal region. Because the system skirted rather than directly crossed Gujarat, the expected impact pattern changed substantially.

ML / structured-data relevance

Excellent case for track-forecast uncertainty, recurvature, ensemble guidance, and false/changed landfall scenarios.

Meteorological timeline

Date / phase	Meteorological development
9 Jun	Low-pressure area formed over the southeast Arabian Sea and adjoining Lakshadweep/eastcentral Arabian Sea.
10 Jun	System organized and intensified into a Cyclonic Storm.
11 Jun	Rapid intensification continued; wind increased substantially.
12 Jun	Reached VSCS strength, with peak intensity maintained during the following period.
13 Jun	Moved close to the Gujarat coast and began to skirt the coast rather than make landfall.
14 Jun	Continued west/northwestward away from the Gujarat coast.
16 Jun	Recurved again toward the Gujarat region while weakening.
17 Jun	Weakened over the Arabian Sea into a well-marked low-pressure area.

Structured-data record

Field	Value
storm_name	VAYU
season	2019
basin_code	NI
track_start	10 Jun 2019

Field	Value
track_end	17 Jun 2019
peak_wind_kt	80
min_pressure_hpa	970
track_length_km	1862

Source set

• IMD/RSMC New Delhi — Very Severe Cyclonic Storm VAYU report

• IMD Annual Report 2019

• NOAA/NCEI IBTrACS Version 4.01 documentation

3. Cyclone TAUKTAE (2021)

Period: 14 – 19 May 2021
Basin: Arabian Sea
Classification: Extremely Severe Cyclonic Storm (ESCS)

Parameter	Details
Genesis	Arabian Sea
Landfall / closest approach	Saurashtra coast, Gujarat, near the Gujarat coast on 17 May 2021
Peak intensity	100 kt (~185 km/h), gusting to 210 km/h
Minimum central pressure	950 hPa
Track length	1880 km
Life period	5 days 9 hours (129 hours)

Research significance

A major Arabian Sea cyclone that affected much of India's west coast before crossing Gujarat.

Impact summary

Heavy to extremely heavy rain, damaging winds and coastal hazards affected Lakshadweep, Kerala, Karnataka, Goa, Maharashtra and Gujarat; remnants also produced heavy rain farther north.

ML / structured-data relevance

Strong example for rapid intensification, coastal impact forecasting, landfall verification and multi-state warning systems.

Meteorological timeline

Date / phase	Meteorological development
14 May	Depression stage marked the beginning of the documented cyclone life cycle.
15 May	Strengthened while moving north/northwest over the Arabian Sea.
16 May	Rapid intensification began; maximum sustained wind increased markedly.
17 May	Reached peak intensity of 100 kt and made landfall on the Gujarat/Saurashtra coast.
18 May	Maintained cyclone-strength impacts over Gujarat for an extended period before weakening.
19 May	Weakened substantially inland; remnants affected northwestern India.
17–18 May	The system produced heavy to extremely heavy rainfall and strong winds across several west-coast states.
Post-landfall	The weakening phase remained relevant for rainfall and flooding analysis beyond the immediate coast.

Structured-data record

Field	Value
storm_name	TAUKTAE
season	2021
basin_code	NI
track_start	14 May 2021

Field	Value
track_end	19 May 2021
peak_wind_kt	100
min_pressure_hpa	950
track_length_km	1880

Source set

• IMD/RSMC New Delhi — Preliminary Report on ESCS TAUKTAE

• IMD Annual Report 2021

• NOAA/NCEI IBTrACS Version 4.01 documentation

4. Cyclone YAAS (2021)

Period: 23 – 28 May 2021
Basin: Bay of Bengal
Classification: Very Severe Cyclonic Storm (VSCS)

Parameter	Details
Genesis	West-central Bay of Bengal
Landfall / closest approach	North Odisha coast, near Balasore, India, on 26 May 2021
Peak intensity	75–85 kt (~140–157 km/h)
Minimum central pressure	About 970 hPa class in the IMD best-track/reporting record
Track length	About 703 km
Life period	About 114 hours at D-to-D stage convention

Research significance

A compact but intense Bay of Bengal cyclone with significant coastal flooding and wind/rain impacts in Odisha and West Bengal.

Impact summary

Odisha and West Bengal experienced strong winds, heavy rainfall and coastal flooding; the system also affected Jharkhand and Bihar after landfall.

ML / structured-data relevance

Useful for studying short-track cyclones, coastal flooding, intensity evolution and forecast-to-impact relationships.

Meteorological timeline

Date / phase	Meteorological development
23 May	A Depression formed over the west-central Bay of Bengal.
23 May	The system strengthened to Deep Depression and then Cyclonic Storm YaaS.
24 May	Intensified to Severe Cyclonic Storm while moving north-northwestward.
25 May	Strengthened to Very Severe Cyclonic Storm.
26 May	Crossed the north Odisha coast as a VSCS.
27 May	Weakened inland while producing rainfall over eastern and central India.
28 May	The system dissipated/weakened substantially after moving inland.
Impact phase	Coastal flooding, heavy rain and damaging winds were the main hazards.

Structured-data record

Field	Value
storm_name	YAAS
season	2021
basin_code	NI

Field	Value
track_start	23 May 2021
track_end	28 May 2021
peak_wind_kt	85
min_pressure_hpa	970
track_length_km	703

Source set

• IMD/RSMC New Delhi — Preliminary Report on VSCS YAAS

• IMD Annual Report 2021

• NOAA/NCEI IBTrACS Version 4.01 documentation

5. Cyclone MOCHA (2023)

Period: 9 – 15 May 2023
Basin: Bay of Bengal
Classification: Extremely Severe Cyclonic Storm (ESCS)

Parameter	Details
Genesis	Southeast Bay of Bengal and adjoining south Andaman Sea
Landfall / closest approach	Between Kyaukpyu, Myanmar and Cox's Bazar/Bangladesh region on 14 May 2023
Peak intensity	110 kt (~204 km/h)
Minimum central pressure	About 938 hPa class during peak intensity in operational best-track reporting
Track length	Approx. 2000 km class
Life period	About 6 days from depression to weakening, depending on stage convention

Research significance

A rapidly intensifying Bay of Bengal cyclone that reached ESCS strength before landfall in the Myanmar–Bangladesh region.

Impact summary

Very strong winds, heavy rainfall, storm surge and coastal hazards affected Myanmar and Bangladesh, with particularly serious humanitarian risk around the Rakhine and Cox's Bazar region.

ML / structured-data relevance

Strong case for rapid-intensification detection, satellite monitoring, track/intensity prediction and early-warning research.

Meteorological timeline

Date / phase	Meteorological development
8 May	Low-pressure area formed over the southeast Bay of Bengal and adjoining south Andaman Sea.
9 May	System concentrated into a Depression.
11 May	Strengthened into Cyclonic Storm MOCHA.
12 May	Intensified into an Extremely Severe Cyclonic Storm.
13–14 May	Reached peak intensity around 110 kt over the east-central Bay of Bengal.
14 May	Made landfall between the Myanmar coast near Kyaukpyu and the Bangladesh region.
15 May	Moved inland over Myanmar and rapidly weakened.
Forecasting note	IMD reported useful pre-genesis and pre-landfall guidance several days ahead of the event.

Structured-data record

Field	Value
storm_name	MOCHA
season	2023
basin_code	NI
track_start	9 May 2023

Field	Value
track_end	15 May 2023
peak_wind_kt	110
min_pressure_hpa	938
track_length_km	2000

Source set

• IMD/RSMC New Delhi — Extremely Severe Cyclonic Storm MOCHA brief report

• IMD Annual Report 2023

• NOAA/NCEI IBTrACS Version 4.01 documentation

How to connect this dossier to the ML validation task

• Filter the IBTrACS North Indian Ocean subset by storm name and year.

• Keep timestamp, latitude, longitude, maximum sustained wind and minimum central pressure.

• Normalize missing values and timestamps before matching observations.

• Use official IMD/RSMC best-track rows as the reference for India-focused evaluation.

• For predicted-versus-actual cyclone-center validation, match timestamps and calculate great-circle distance with the Haversine formula.

• Report MAE in kilometres only after the actual prediction CSV and observed best-track CSV are available.

Recommended data schema

storm_name	season	basin	datetime_utc	latitude	longitude	max_wind_kt	min_pressure_hpa	source

Important: Do not calculate a real forecast MAE from this narrative report alone. The Day-4 metric requires actual predicted and observed timestamp-level positions.